# Observer Patch Holography Cosmology Data and Likelihood Contracts

Provenance, Comparability, and Claim Boundaries

B. Müller

November 4, 2026

**Paper release:** `r2061` **Released:** November 4, 2026

**Author affiliation:** Bernhard Mueller, Pragma Research Inc.

**Abstract**

A cosmological comparison requires a finite source model, compatible transfer, an observable projection and a data likelihood with covariance and nuisance parameters. We distinguish quantities derived from a chosen source from imported scales, fitted parameters and diagnostic formulas. A reproducible retrospective ledger compares conditional observer-patch values with Planck, ACT, SPT, BICEP/Keck, DESI and SPARC measurements. It records both agreement and tension, including the conditional tilt, fixed-capacity equation of state and capacity normalization. These comparisons use seen data and several share the standard cosmological baseline. None supplies a joint source-derived cosmological likelihood or a likelihood ratio for the full theory.

## 1 Comparison Classes

Every reported number belongs to one of four classes:

1. **source-derived**: all physical inputs follow from the declared OPH construction without using the comparison data;
2. **imported**: established background or microphysical inputs are declared explicitly;
3. **fit**: one or more parameters are inferred from the same data;
4. **diagnostic**: the calculation tests shape, scale, code behavior, or an interface without supporting a physical OPH claim.

The cosmology comparisons in scope are diagnostic or imported. No source-derived joint cosmology likelihood exists. The retrospective comparison ledger of Section 5 classifies every recorded comparison by these classes and by whether the compared value is shared with the standard baseline.

In a likelihood comparison, the event law of the spacetime construction [1], its cone and count-clock hypotheses and their physical calibration count as model information, alongside the background, conserved stress, initial conditions and matter assumptions. A geometric count clock by itself does not define the cosmological transfer or observable likelihood.

## 2 Model and data contract

A comparison must specify the covariant action and source map, background and species, gauge-invariant initial data, perturbation and collision terms, observable projection, numerical errors, and every external input or fitted parameter. Source, transfer and observable must use the same physical clock and scale.

It must also identify the data release, selection, masks, covariance, calibration, nuisance parameters, priors and combination rule. Sample overlap and shared calibration enter the covariance; a sum of diagonal residuals is not the joint likelihood when correlations are present. A measured quantity that selects the formula, normalization or applicability range is an input to that comparison. For example, conserved modular charge in a comoving volume gives $\rho_A \propto a^{-3}$, and with a separate no-exchange continuity law gives a pressureless background. It does not fix abundance, perturbations, lensing or nonlinear growth.

## 3 Observable Contracts

| Observable family | Required theory output | Required comparison object |
|---|---|---|
| Cosmic microwave background (CMB) | $C_\ell^{TT}$, $C_\ell^{TE}$, $C_\ell^{EE}$, lensing, foreground model | map or band-power likelihood with covariance and nuisance model |
| Baryon acoustic oscillations (BAO) and supernovae | $H(z)$, $D_A(z)$, luminosity distance, sound horizon | survey likelihood with calibration covariance |
| Weak lensing | metric potentials, nonlinear matter power, intrinsic alignments | tomographic likelihood with masks and covariance |
| Growth and redshift-space distortions (RSD) | $P_m(k,z)$ and $f\sigma_8(z)$ | windowed survey likelihood with cross-bin covariance |
| Clusters | mass function, selection, observable–mass map, lensing calibration | count and calibration likelihood |
| Galaxies | disk dynamics, gas and stellar nuisances, external environment | galaxy-level likelihood with selection and covariance |

## 4 Diagnostic Results

The reported CMB overlays, compressed background comparisons, and galaxy scaling checks are diagnostic. Their formulas or inputs are selected with knowledge of the comparison data, omit required covariance or nuisance structure, or lack a physical source map. They test numerical paths and conditional formulas and do not establish a cosmological prediction.

**Conditional observer spectra and native-source tests.** The observer-spectrum calculation propagates a supplied primordial amplitude and shape through a supplied cosmological background and standard temperature/polarization transfer. A conditional sky realization adds cosmic-variance sampling around that mean; it does not select the mean's inputs. Agreement with a measurement-calibrated reference curve, including equality when identical source and transfer inputs are used, verifies a numerical interface and cannot be counted as independent source evidence. Residual root mean squares formed from marginal band-power errors omit inter-bin covariance and nuisance parameters; they are not a likelihood, probability of the theory, or independent fit significance.

The native-repair tests have a different observable and evidence source: absolute stationary load-record variance and inherited/fresh/settled covariance in 61 retained refinement chains. They constrain those finite mechanisms without using a CMB normalization, but do not identify load as geometric curvature, set a physical background, or supply a primordial tilt. In the executable observer construction, actual native density snapshots on refining source screens have joint Gaussian limits with covariance $\exp(-|\tau - u|A)$ for a positive operator $A$ of spectral dimension two on the screen, with $\tau$ a rescaled native attempt count [4, 3]. That construction does not derive the tilt branch $n_s = 1 - P/48$ from its joint records [5]. Their receipts and conditional observer comparisons are archived [7]. An observational test must first connect the native geometric readout to the source, radial and transfer interfaces with their uncertainties.

## 5 Retrospective Comparison Ledger

The deterministic cosmology ledger records the conditional comparisons in Table 1. Its producer reads the corpus values and the parent receipts, types each public measurement once with its citation, and writes one row per comparison with a class label; an independent verifier recomputes every row. Every row is on seen data. No row is a frozen prediction or a score, and the rule, theory minus measurement over the quoted standard deviation, is fixed before any row is read. Table 1 summarizes the rows.

| Quantity | Value | Public measurement and distance | Class |
|---|---|---|---|
| $n_s$ | $1 - P_\star/48 = 0.96602$ | Planck 2018 $0.9649 \pm 0.0042$ ($+0.27\sigma$); SPT-3G with Planck $0.9636 \pm 0.0035$ ($+0.69\sigma$); SPT, Planck and ACT $0.9684 \pm 0.0030$ ($-0.79\sigma$); ACT DR6 with Planck $0.9709 \pm 0.0038$ ($-1.28\sigma$); with lensing and DESI 2024 (DR1) BAO $0.9743 \pm 0.0034$: lower 0.612 percent posterior position in the official chains (Gaussian summary $-2.43\sigma$) | declared-density value at the calibrated grain, like the capacity display |
| $\mathrm{d}n_s/\mathrm{d}\ln k$ | 0 | Planck 2018 $-0.0045 \pm 0.0067$ ($+0.67\sigma$); ACT DR6 with Planck, lensing and BAO $0.0062 \pm 0.0052$ ($-1.19\sigma$) | conditional theorem; discriminates only against running of order $10^{-2}$ |
| $r$ | 0 | BICEP/Keck 2018 $r_{0.05} < 0.036$; the fourth Planck release with BK18 and BAO $r < 0.032$; consistent | requires zero tensor initial data and forcing; tests specified alternatives |
| $\beta_{\mathrm{iso}}$ | 0 | Planck 2018 uncorrelated CDI $100\beta_{\mathrm{iso}} < 3.5,\ 3.8,\ 3.9$ at the three reference scales; consistent | shared baseline |
| $f_{\mathrm{NL}}$ | 0 at the source | Planck 2018 local $-0.9 \pm 5.1$ ($+0.18\sigma$), equilateral $-26 \pm 47$, orthogonal $-38 \pm 24$ | shared baseline |
| $\Omega_K$ | 0 | Planck 2018 with lensing $-0.0106 \pm 0.0065$ ($+1.63\sigma$); with BAO $0.0007 \pm 0.0019$ ($-0.37\sigma$); DESI DR2 with CMB $0.0023 \pm 0.0011$ ($-2.09\sigma$) | shared baseline |
| $(w_0, w_a)$ | $(-1, 0)$ | DESI DR2 BAO with CMB $2.61\sigma$; with Pantheon+ $2.60\sigma$; with Union3 $3.38\sigma$; with DESY5 $3.93\sigma$ (Gaussian moment diagnostic of the official chains) | conditional theorem; tension |
| $\Lambda\ell_P^2$ | $3\pi/N$, $N = 3.29$ and $3.30 \times 10^{122}$ | Planck-central $\Lambda$: $+0.62$ and $+0.38$ percent; DESI DR2 with CMB display $(2.968 \pm 0.040) \times 10^{-122}$: $-3.5$ and $-3.8$ percent ($-2.6$ and $-2.8\sigma$) | closure-candidate display |
| $a_0$ | no source-derived value | Registered readings 1.1847 and $1.1790 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. SPARC calibration $1.161 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at mass-to-light 0.5; ratio to $c^2\sqrt{\Lambda/3}$ equal to 0.214 | fitted comparison value |

Table 1: Rows of the cosmology postdiction ledger. Distances are theory minus measurement in units of the quoted standard deviation; bounds are consistent when the value lies below the bound. All data are seen. SPARC values are ineligible for scoring the registered acceleration readings.

Agreement has to be attributed to the tested model. Zero source isocurvature and Gaussian source cumulants are shared with standard adiabatic models; $(w_0, w_a) = (-1, 0)$ is the ΛCDM point. The tensor-zero branch additionally assumes zero initial tensor amplitude and velocity and no forcing. Its compatibility with a finite upper bound does not establish exact zero or exclude all slow-roll models. The running-zero branch discriminates only alternatives resolvable by the data.

The tilt uses a declared full-collar generator density $P_\star/24$, evaluated at a pixel located from the measured fine-structure constant. No native source emits that density. A conditional particle/memory root also does not derive it. In the ACT DR6, Planck, lensing and DESI joint chains, 0.612 percent of posterior mass lies below 0.96602, versus approximately 0.74 percent in a Gaussian approximation; the diagnostic 0.96484 branch lies at 0.200 percent [2, 5]. These are posterior positions conditional on the fitted baseline, not probabilities of OPH or likelihood ratios.

The fixed-capacity point has a 2.6 to $3.9\sigma$ Gaussian-moment distance in the stated DESI combinations, and the monotone-capacity branch has posterior fractions between 0.007 and 0.11 percent. The two capacity displays differ from the DESI coordinate by $-2.64$ and $-2.81\sigma$, while differing from the Planck-central value by $+0.62$ and $+0.38$ percent. All these inputs were seen before the registered comparison protocols. Only a complete prospectively fixed protocol, its eligible release, statistic and thresholds defines a test; the frozen prediction register specifies those separate contracts [6]. A failure of a conditional branch rejects the full proposal only if that branch's premises are shown necessary.

## 6 No-Data-Use Boundary

A source-derived calculation must expose its dependency graph. The source side may use mathematical constants, declared OPH finite artifacts, and explicitly imported physical inputs. It may not use the observed quantity being claimed as an output, a fit to that quantity, or a proxy calibrated from the same dataset.

This provenance rule is enforced through the declared dependency graph and data-use classification.

## 7 Theorem scope

The screen-spectrum and radial-lift mathematics are conditional theorems. One-shell inversion is nonunique; physical source dilation and radial cross-covariance tomography are two established sufficient uniqueness routes. Neither route is supplied by the native screen energy, which is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk; a radial lift needs a radial observation algebra as an added premise [3]. Finite source instantiation, repair-charge cosmological source, Boltzmann bridge, and joint likelihood are not supplied. All cosmological data products are diagnostic. No cosmological result carries an OPH falsification verdict.

## References

[1] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/p

aper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf.

[2] T. Louis et al. (2025). The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters. arXiv:2503.14452. https://arxiv.org/abs/2503.14452.

[3] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/model/physical_space_time/OCCUPATION_FIELD.md.

[4] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/model/observer_time_fields/REFINEMENT.md.

[5] FloatingPragma (2026). OPH predictions and public-data comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md.

[6] FloatingPragma (2026). Frozen prediction register. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/claims/frozen_prediction_register.json.

[7] FloatingPragma (2026). Observer-dynamics analysis archive: native-repair receipts and conditional observer comparisons. https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/observer_dynamics_20260925/sim-analysis/codex.

## AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.